

AZURE CLOUD DEPLOYMENT

Taking AI to the Web

Moving the College FAQ Bot from Local to Live Azure Cloud Service

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The Vision & Motivation

Ensuring instant campus information accessibility for students at
Nagarathinam Angalammal Arts and Science College.

Information Accessibility Gaps

The Problem

Campus information is often scattered across PDFs, notice boards, and manual queries, leading to delays and student confusion during crucial academic hours.

The AI Solution

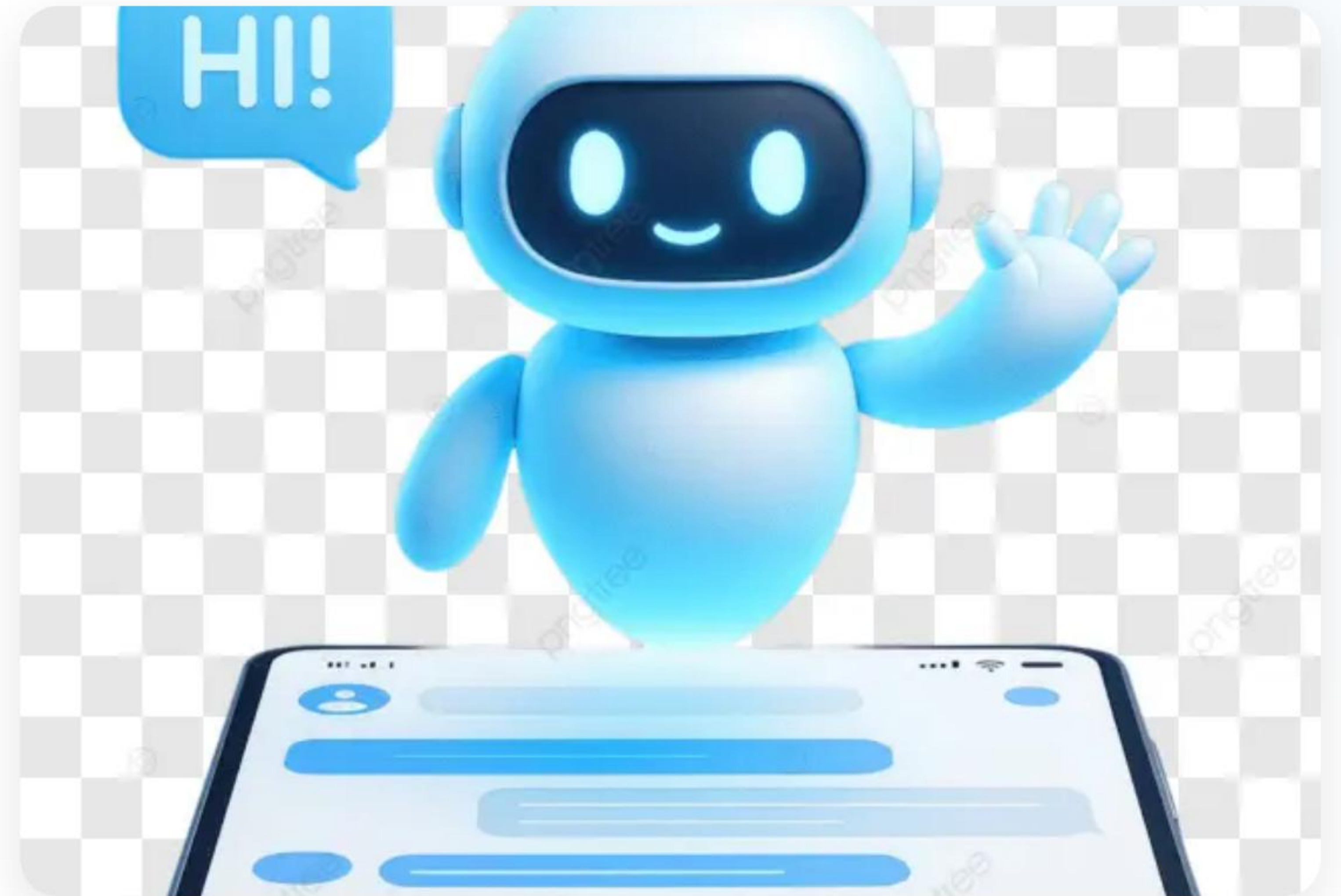
A centralized AI-powered FAQ bot that provides instant, verified answers 24/7, hosted on the web for cross-device accessibility.

Interactive AI Experience

Real-World Usability

The bot handling real campus-related questions in a web browser. It provides a conversational interface where students can ask about admissions, course details, and faculty information.

Impact: Students no longer need to wait for office hours. The bot brings the college office to their smartphones.



Technology Stack



Azure App Service

Hosting the bot's interface and managing the cloud environment for stable performance.



AI Logic

Processing complex queries and transforming text into usable campus applications.



Web Hosting

Ensuring the bot stays fast and reliable for users across any web browser.

From Local to Live

Deployment on the web taught me about the bridge between AI models and the end-user. Moving away from localhost to a live URL was a major milestone.

Cloud Insights:

- Environment configuration
- App scaling strategies
- Secure cloud hosting

Key Learning Outcomes

- ✓ **Azure Management:** Mastered the hosting of web app services and cloud environment management.
- ✓ **Real Interaction:** Observed how AI models behave when exposed to real student query variations.
- ✓ **Scalability:** Learned how the Azure backend ensures reliability during high traffic peaks.
- ✓ **Building in Public:** Understanding that perfection is secondary to usability and iteration.

Deployment Efficiency

99%

Availability Target

Fast & Reliable

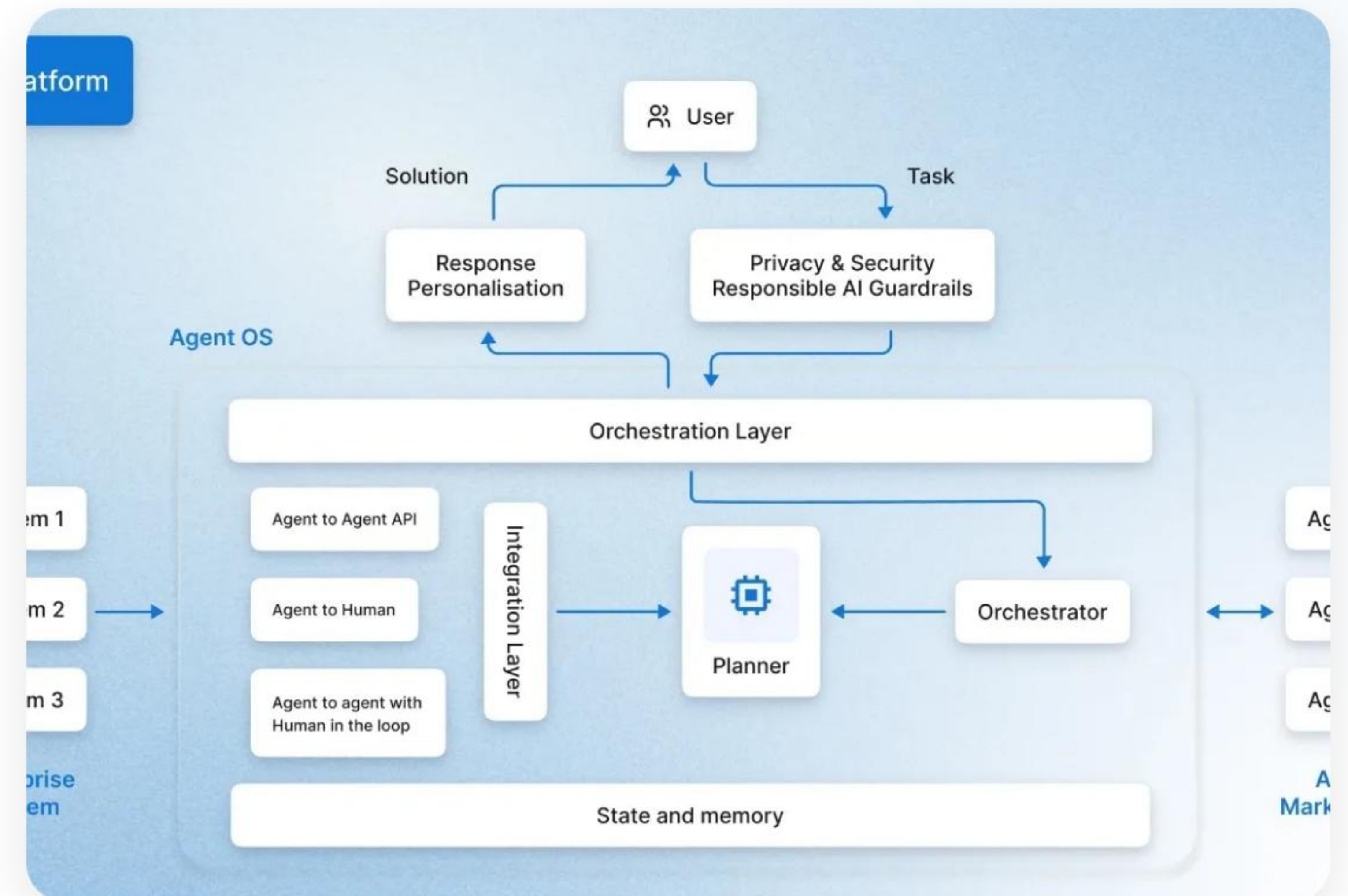
By deploying on Microsoft Azure, the bot infrastructure guarantees high uptime and low latency. This ensures that when a student needs information urgently, the bot is always ready to respond.

Next Step: Agentic Logic

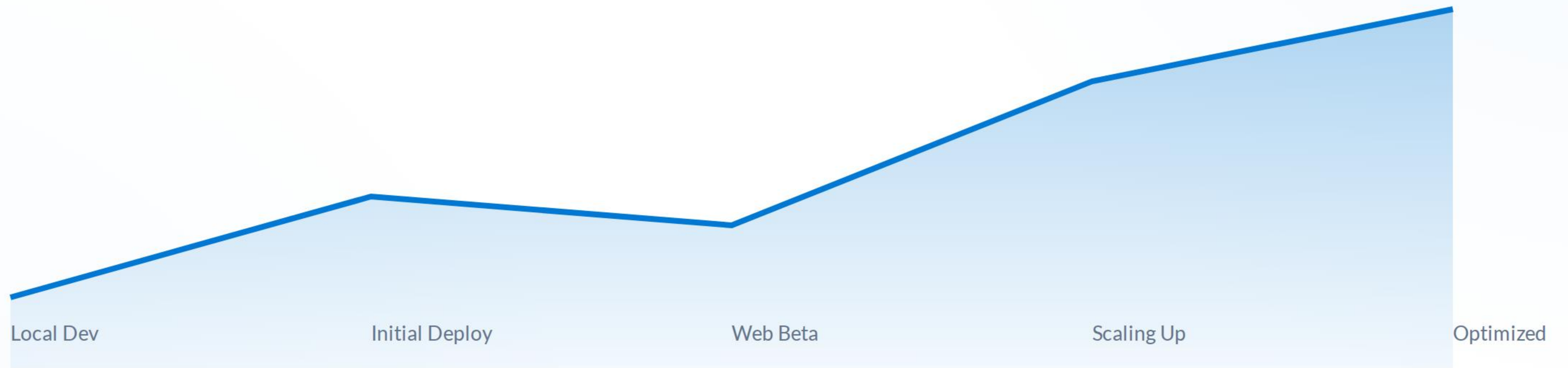
Future Roadmap

Integrating Mastra 2.0 patterns to make the bot more agentic, handling more complex multi-step student queries.

$$\text{Agent}_{t+1} = f (\text{Query}_t + \text{Mastra}_{2.0})$$



Traffic Response Trends



Responsiveness and reliability growth across deployment phases.

Building in Public



Growth Mindset

“It’s not perfect yet, but that’s the beauty of building in public! Seeing it live on a URL is a huge milestone.”

Collaborating with mentors and peers helps refine the agent logic and explore new patterns to handle student needs better.

Thank You!

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 github.com/karthickr

Image Sources



https://png.pngtree.com/png-vector/20251210/ourlarge/pngtree-cute-ai-chatbot-waving-above-smartphone-with-message-interface-png-image_18146766.webp

Source: [pngtree.com](https://png.pngtree.com/)



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Source: www.kore.ai



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